Jung Hyun Bae

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EDUCATION

Aug. 2024 – **Georgia Institute of Technology, Atlanta, Georgia**
Doctor of Philosophy in Robotics

Aug. 2022 – Aug. 2024 **University of Texas at Austin, Austin, Texas**
Master of Science in Mechanical Engineering (Dynamic Systems and Controls Track)

Mar. 2015 – Feb. 2022 **Chung-Ang University, Seoul, South Korea**
Two years of military service
Bachelor of Science in Mechanical Engineering

RESEARCH EXPERIENCE

Jun. 2025 – Aug. 2026 Department of Anesthesiology, Emory University, Atlanta, GA

- **Research Project: BQST – Benchmarking Physical Accuracy of Quantitative Sensory Testing (QST) Devices with a Human-Like Skin Phantom**
  - Designed and fabricated a human-like gelatin skin phantom to establish physical accuracy benchmarks for quantitative sensory testing (QST) devices used in pain assessment.
  - Characterized the delivery accuracy of thermal and mechanical QST stimuli against the phantom to support standardized device calibration for pain research.

Oct. 2024 – Sep. 2025 Uncommon Sense Labs, Georgia Institute of Technology

- **Research Project: µMobileScan – Smartphone Whole-Slide Imaging for Red Blood Cell Counting**
  - Co-developed a smartphone whole-slide imaging application with real-time capture guidance to automate red blood cell counting for low-cost, point-of-care hematology.

Aug. 2024 – May. 2025 Uncommon Sense Labs, Georgia Institute of Technology

- **Research Project: Inductus – A Soft-Sensing System for Fetal Health Monitoring**
  - Developed a high-resolution soft-sensor system to continuously monitor fetal mechanical activity in utero, extending fetal health tracking beyond heart rate.
  - Validated the system in vitro with a fetal phantom, demonstrating accurate detection of fetal movement through vibration sensing.

Aug. 2024 – Feb. 2025 Uncommon Sense Labs, Georgia Institute of Technology

- **Research Project: LumiBite – Personalized Bottom-up Lighting Lunchbox**
  - Contributed to an in-the-wild technology probe investigating a personalized bottom-up lighting lunchbox designed to enhance the dining experience.

Sep. 2022 – Aug. 2024 Rehabilitation and Neuromuscular Robotics Lab, University of Texas at Austin

- **Research Project: ACT-Hand Design and Optimization of Nonlinear Control Algorithm**
  - Designing Anatomically Correct Testbed (ACT) simulating the musculoskeletal structure of the fingers of the human hand
  - Integrated Series Elastic Actuators (SEAs) into the ACT hand to enable active stiffness adjustment, enhance compliance under nonlinear constraints, thereby achieving precise end-tip stiffness control.

Aug. 2023 – Dec.2023 Human Centered Robotics Lab, University of Texas at Austin

- **Research Project: PLATO; Robot hand manipulation and teleoperation**
  [Research Support: Sony Corporation]
  - Planning and simulating motion tasks using MoveIt and the Gazebo environment for

the robot hand manipulation.

Jan. 2023 – Jul. 2023 Precision Mechatronics and Control Lab, University of Texas at Austin

- **Research Project: "A Novel Soft Electromagnetic Actuator with Lorentz-type Design and Ferrofluidic Bearings for High Strength, Efficiency, and Transparency"**
  [Research Support: Meta Platforms, Inc.]
  - Developed a novel soft electromagnetic actuator for wearable haptic devices, tailored to maximize ferrofluid functionality, thereby improving performance.

Sep. 2022 – Apr. 2023

- **Research Project: Design, Fabrication and Modelling of Two Bio-inspired Rigid-soft Coupling Wearable Pneumatic Robots**
  - Designed and fabricated a two-degree-of-freedom soft robot system, including a silicone pneumatic actuator

Sep. 2020 – Jun. 2021 Assistive and Rehabilitation Robotics Lab, Chung-Ang University, Seoul, Korea

- **Research Project: "FLEX: Development of Smart Wearable Suit Combined with Deep Learning for Muscle Power Assistance, Injury Prevention, and Work Efficiency Enhancement for Forest Workers"**
  [Research Support: Korea Forest Service]
  - *Research Assistant, Mechanical Design team (Actuator Structure Design)*
    - Designed an actuator used on soft exosuit for reducing the metabolic cost
    - Enhanced the performance of the actuator by introducing a two-way operable dual pulley mechanism using the moving gear capable of automatic dislocation
    - Optimized the heat treatment of related parts including torsional spring based on quenching-tempering process
  - *Research Assistant, Electronics team (Motor Control)*
    - Designed electronic system mounted on the exosuit

RESEARCH SKILLS & CERTIFICATES

Design Software	Solidworks, Autodesk Inventor, KiCad
System Control	Python, C++, ROS 2, MATLAB, LabVIEW
Analysis Software	Ansys, Minitab, Abaqus
Simulation/Visualization	Gazebo, Visual3D(Biomechanical Analysis of Visual 3D), Blender, Premiere Pro, Photoshop
Certificates	MENSA Korea membership BAV(Biomechanical Analysis of Visual 3D) Level II Adult and Pediatric First Aid/CPR/AED, American Red Cross Fire Safety Management Level 2, Korea Fire Safety Institute

PUBLICATION

Jung Hyun Bae, Timothy Jun, Sixuan Wu, Daniel Harper, Alexander Adams. (2026). Establishing Physical Accuracy Benchmarks for QST Devices Using a Human-Like Skin Phantom. The Journal of Pain, United States Association for the Study of Pain (USASP).

Sixuan Wu, **Jung Hyun Bae**, Alexander T. Adams. (2026). µMobileScan: A Smartphone Whole Slide Imaging App for Red Blood Cell Counting with Real-time Guidance. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT).

Haiqing Xu, Xiwen Yao, Sixuan Wu, **Jung Hyun Bae**, Zhifan Guo, Dian Lv, Zhihao Yao, HyunJoo Oh, Alexander T. Adams. (2026). LumiBite: An In-the-Wild Technology Probe Exploring Personalized Bottom-up Lighting Lunchbox for Enhanced Dining Experiences. Proceedings of the ACM International Conference on Tangible, Embedded, and Embodied Interaction (TEI '26).

Haiyun Zhang, Gabrielle Naquila, **Jung Hyun Bae**, Zonghuan Wu, Ashwin Hingwe and, Ashish Deshpande. (2024). Novel bio-inspired soft actuators for upper-limb exoskeletons: design, fabrication and feasibility study. Frontiers in Robotics and AI.

Jung Hyun Bae. (2024). Development of the ACT Hand with a Broad and Precisely Adjustable Stiffness Spectrum via Series Elastic Actuators. Graduate Thesis Paper for M.S. degree, University of Texas at Austin(UTA).

Jung Hyun Bae. (2021). Designing Wire-driven Running Reinforcement Shoes. Graduate Thesis Paper for B.S. degree, Chung-Ang University (CAU).

Jung Hyun Bae, Myunghwan Song. (2021). Design of LED Rubber Cone for Triboelectrification-Based Power Generation. Graduate Thesis Paper for B.S. degree, Chung-Ang University (CAU).

WORK EXPERIENCE

Jan. 2025 – Aug. 2026 **Georgia Institute of Technology**
Teaching Assistant (Health Sensing & Interventions)
- Assisted in a health technology course by supporting student projects, providing feedback on proposals, and guiding IRB certification completion.

Aug. 2024 – Dec. 2024 *Teaching Assistant (Introduction to Graduate Studies)*
- Provided detailed, constructive feedback on research proposals, guiding students in identifying innovative research topics, conducting effective literature reviews, and employing proper academic citation practices.

Aug. 2023 – Aug. 2024 **The University of Texas at Austin**
Teaching Assistant (Dynamic System & Controls Lab)
- Directed experiments on mechanical vibration systems using Arduino and Python, complementing theoretical instruction with data measurement and analysis

Sep. 2017 – Aug. 2019 **VHS(Veterans Health Service) Medical Center, Seoul, Korea**
Facility Equipment Manager
- Repair and calibration of medical equipment, such as Non-Invasive Blood Pressure devices

May. 2017 – Jun. 2017 **KWANGGEON T&C, Chungcheongbuk-do, Korea**
- Constructed and supplied materials for building interior of SK Hynix Semiconductor Industrial Complex

ACADEMIC EXPERIENCE

Mar.2021 - Jun. 2021 Capston Design Project, Mechanical Engineering, Chung-Ang University
- **Project Title: Wire-driven Running Reinforcement Shoes**
  - Designed wearable running reinforcement device by using Solidworks, Ansys
  - Utilized planetary gear system and wire-driven mechanism to deliver the energy from running motion's arm movement to soles to strengthen the running propulsion
  - Worked on a graduation thesis (Completed advisor review)

Jan. 2021 – Jan. 2021 **Certified BAV(Biomechanical Analyst of Visual 3D) Level II, Vector Biomechanics**
- Completed 18-hour course on biomechanics software Visual 3D
- Utilized Visual 3D for advanced biomechanical analysis, including Full-Body Modeling and EMG Data Processing of walking motions

Jul. 2020 – Aug. 2020 **Mechanical Engineering Short Term Research Program, Chung-Ang University**
- Introduced a belt mechanism on the high-power actuator for wearable robot to develop a power transmission system with lighter weight and smaller size

Oct. 2018 – Oct. 2018 **"Samsung Medical Center" & "Digital Healthcare Partners(DHP)" & "Samsung Advanced Institute for Health Sciences & Technology (SAIHST)" & "Seoul Center for Creative Economy & Innovation", Seoul, Korea**
- Used JavaScript to make "Healthcare Service Program" that uses AI voice recognition system to provide health information to patients at hospital

AWARDS, SCHOLARSHIPS AND HONORS

15. Oct. 2020	**2020 Summer Internship**(Mechanical Engineering Short Term Research Program; MESTER) Excellence Award, Mechanical Engineering Department, Chung-Ang University
Fall Semester. 2019	**Scholarship** (Academic Excellence), Chung-Ang University, Seoul, South Korea
05. Dec. 2018	**Award for Excellent Volunteer Service**, (100 hour), Yangcheon-gu Volunteer Center
01. Sep. 2017	**Citation Award**, Veterans Health Service Medical Center
Fall Semester. 2016	**Scholarship for Volunteer Work**, Mechanical Engineering Department, Chung-Ang University

VOLUNTARY/ EXTRACURRICULAR ACTIVITIES

Aug. 2021 (24 Hours) **C-Design Thinking Academy, Central Libary, Chung-Ang University, Seoul, South Korea**

- Worked on the design thinking process for finding and coming up with creative problem-solving methods in multidisciplinary aspects

Feb. 2021 – Apr. 2021 **PTStar; Speech and Presentation Club, Seoul, South Korea**

- Training on presentation planning, team presentation, and one-minute speech

Feb. 2020 – May. 2020 (12 Hours) **'SNS(Saturday & Sunday Neighborhood Service)' Voluntary Club, Seoul, South Korea**

Feb. 2019 – Apr. 2019 **DEMA Studio, Seoul, South Korea**

- Engaged in biweekly discussions on "Design Thinking" and activity applied the principles to address real-world challenges, fostering the development of creative and innovative thinking skills

Mar. 2015 – Jan. 2019 (87 Hours) **Child Care and Education Volunteer Club, Chung-Ang University, Seoul, South Korea**

- Guidance and instruction in academic learning for foster children

Mar. 2018 - Dec. 2018 (92 Hours) **KT&G Cooperative Volunteer Work Club of Yangcheon-gu Volunteer Center, Seoul, Korea**
Student President